

Annexure III: Teaching Strategy

CLO No.	Teaching-Learning Activities for Active Learning	Assessment (Formative and Summative) Activities
CLO 1	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Practical sessions	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 2	1. Discussion 2. Reflection 3. Assignment on topics 4. Practical Sessions 5. Think Pair Share	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 3	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Practical Sessions	1. Continuous Assessment 2. Practical exam 3. Assignment
CLO 4	1. Discussion 2. Problem Solving 3. Assignment on topics 4. Practical Sessions	1. Continuous Assessment 2. Practical exam 3. Assignment

Annexure IV: Course Outline

Institute	Institute of Technology
Department	Computer Science and Engineering
Programme	B.Tech.
Batch	2024-2028
Term/Semester	5 th
Course Title	Full Stack Development
Course Code	3CS524CC26
Credit Hours	L-3+P-2
Faculty's Name	Dr Smita S Agrawal Dr Devendrasinh Vashi, Dr Sunil Gautam, Dr Rameshram Naik Dr Sanjay Patel Prof. JJP , Prof. MR, Prof. BBS,
E-mail ID	smita.agrawal@nirmauni.ac.in
Contact No.	079-71652539
Office Hours	Mon to Fri: 07:30 am to 3:00 pm Sat: 7:30 am to 1:30 pm
Moodle Details (as and when available)	https://lms.nirmauni.ac.in/course/view.php?id=10674

a. Course Overview (300 to 1000 words)

Full Stack Development refers to the comprehensive set of skills and technologies required to develop both the front end (client-side) and back end (server-side) of a web application. A Full Stack Web Developer is proficient in various programming languages, frameworks, and tools, enabling them to work on all aspects of web development.

On the front end, Full Stack Developers focus on creating the user interface and experience. They use languages such as HTML, CSS, and JavaScript, along with libraries and frameworks like React, Angular, or Vue.js, to build responsive and visually appealing interfaces that interact seamlessly with users.

On the back end, Full Stack Developers manage the server-side logic and databases. They work with server-side languages like Node.js, Python, Ruby, or Java, and utilize frameworks such as Express, Django, or Flask. Database management is a crucial aspect, involving interactions with systems like MySQL, MongoDB, or PostgreSQL.

Full Stack Developers also need to be familiar with version control systems like Git, which helps in tracking changes and collaborating with other developers. They often deploy applications on cloud platforms like AWS, Azure, or Google Cloud.

In addition to technical skills, Full Stack Web Developers should understand web architecture, security, and optimization techniques. They need to ensure the application's performance, scalability, and responsiveness across different devices.

In summary, Full Stack Web Development encompasses a diverse skill set, ranging from front-end technologies and design principles to back-end development and database management. This holistic approach allows developers to create end-to-end solutions, making them valuable contributors to the entire web development process

b. Course Learning Outcomes (CLOs) [CLO sequence can be reordered]

At the end of the course, students will be able to –

1. demonstrate the creation of REST APIs for various backend database functionalities of an application (BL2)
2. compare the approaches used for web application development and identify the various components of it (BL4)
3. develop user-friendly and responsive user interfaces (BL6)
4. design end-to-end web applications with various tools and frameworks. (BL6)

c. Text Book (reflect other study/reference materials in session plan)

1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress
2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
3. David Flanagan, JavaScript: The Definitive Guide, O'Reilly
4. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt.
5. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB
6. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace
7. Hugo Di Francesco, Professional JavaScript, Packt
8. Shama Hoque, Full-Stack React Projects, Packt

d. Assessment Components & Schedule

SEE	CE	LPW
0.4	0.3	0.3

Component (with No. of sub-components)	Weightage (each part of the Component)	Schedule	Marks	CLO Number
Class Test	20% of CE	3 rd /4 th week of semester	20	1, 2, 3 and 4
Sessional Test	60% of CE	08-09-2026 to 11-09-2026	50	1, 2, 3 and 4
Innovative Project	20% of CE	1-Aug-2026 to 1-Oct-2026	20	1, 2, 3 and 4
SEE	100% of SEE	19-Nov-2026 onwards	100	1, 2, 3 and 4
LPW Exam	40% of LPW Exam	23-Oct-2026 onwards	40	1, 2, 3 and 4
Practical Work	60% of LPW	6-Jul-2026 to 22-10-2026	100	1, 2, 3 and 4

e. Session Plan

Sess. No.	Topic/Sub Topic	Session Details	
1	JSON and its usage in web applications	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Open discussion, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of JSON and its usage in web applications
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
2	JSON and its usage in web applications (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of JSON and its usage in web applications (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
3	RESTAPIs	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress

			2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Chalk and board lecture, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of REST APIs
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
4	REST APIs (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of REST APIs (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
5	3-Tier Applications, Various Stacks usage for web application development	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB

			4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of 3-Tier Applications, Various Stacks usage for web application development
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
6	3-Tier Applications, Various Stacks usage for web application development (Case Study)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Case study discussion, Open discussion, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of 3-Tier Applications, Various Stacks usage for web application development (Case Study)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
7	Responsive Web Design	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT) with program

		CLO	1,2
		SLO	Understand the concepts of Responsive Web Design
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
8	Responsive Web Design (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of Responsive Web Design (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
9	Media Queries	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of Media Queries
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
10	Media Queries (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress

			2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of Media Queries (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
11	HTML5 Features	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of HTML5 Features
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
12	HTML5 Features	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB

			- Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace - Hugo Di Francesco, Professional JavaScript, Packt - Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of HTML5 Features
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
13	Bootstrap	Text Book	- Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress - Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	- David Flanagan, JavaScript: The Definitive Guide, O'Reilly - Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. - Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB - Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace - Hugo Di Francesco, Professional JavaScript, Packt - Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, Power Point Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of Bootstrap
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
14	Bootstrap	Text Book	- Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress - Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	- David Flanagan, JavaScript: The Definitive Guide, O'Reilly - Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. - Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB - Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace - Hugo Di Francesco, Professional JavaScript, Packt - Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	1,2

		SLO	Understand the concepts of Bootstrap
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
15	Tailwind and CSS based web development	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of Tailwind and CSS based web development
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
16	Tailwind and CSS based web development (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of Tailwind and CSS based web development (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
17	Vue JS: Directives and Binding	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress

		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of Vue JS: Directives and Binding
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
18	Vue JS: Directives and Binding (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of Vue JS: Directives and Binding (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
19	React JS: Elements, UI Components, JSX, Redux, Fetch Data	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace

			5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	1,2
		SLO	Understand the concepts of React JS: Elements, UI Components, JSX, Redux, Fetch Data
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
20	React JS: Elements, UI Components, JSX, Redux, Fetch Data (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of React JS: Elements, UI Components, JSX, Redux, Fetch Data (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
21	Asynchronous Programming, GraphQL APIs	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	1,2

		SLO	Understand the concepts of Asynchronous Programming, GraphQL APIs
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
22	Asynchronous Programming, GraphQL APIs (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	1,2
		SLO	Understand the concepts of Asynchronous Programming, GraphQL APIs (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
23	Java Spring MVC	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Java Spring MVC
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
24	Java Spring MVC (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress

		Reading (Reference) 1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO 4
		SLO Understand the concepts of Java Spring MVC (Hands-on Practice)
		URL https://lms.nirmauni.ac.in/course/view.php?id=10674
25	Spring Boot Framework and its Features	Text Book 1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference) 1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO 4
		SLO Understand the concepts of Spring Boot Framework and its Features
		URL https://lms.nirmauni.ac.in/course/view.php?id=10674
26	Spring Boot Framework and its Features	Text Book 1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference) 1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB

			4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Spring Boot Framework and its Features
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
27	Node JS: Express Framework, Routing and Package Manager	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Node JS: Express Framework, Routing and Package Manager
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
28	Node JS: Express Framework, Routing and Package Manager	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4

		SLO	Understand the concepts of Node JS: Express Framework, Routing and Package Manager
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
29	Application Development with Built-in Modules and Plugins Integrations	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Application Development with Built-in Modules and Plugins Integrations
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
30	Application Development with Built-in Modules and Plugins Integrations	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Application Development with Built-in Modules and Plugins Integrations
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
31	RDBMS - MySQL	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress

			2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3
		SLO	Understand the concepts of RDBMS - MySQL
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
32	RDBMS - MySQL	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3
		SLO	Understand the concepts of RDBMS - MySQL
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
33	NoSQL -- MongoDB	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB

			4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3
		SLO	Understand the concepts of NoSQL -- MongoDB
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
34	NoSQL -- MongoDB	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3
		SLO	Understand the concepts of NoSQL -- MongoDB
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
35	CRUD Operations and Integration with Front End Tools	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3

		SLO	Understand the concepts of CRUD Operations and Integration with Front End Tools
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
36	CRUD Operations and Integration with Front End Tools	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	3
		SLO	Understand the concepts of CRUD Operations and Integration with Front End Tools
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
37	Version Control and Project Management with Git	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Version Control and Project Management with Git
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
38	Version Control and Project	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress

	Management with Git		2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with Java program
		CLO	4
		SLO	Understand the concepts of Version Control and Project Management with Git
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
39	Authentication & Authorisation	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Authentication & Authorisation
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
40	Authentication & Authorisation (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB

			4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	4
		SLO	Understand the concepts of Authentication & Authorisation (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
41	Securing, Deploying and Testing the Application	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Recap lecture, Open discussion, PowerPoint Presentation (PPT) with program
		CLO	4
		SLO	Understand the concepts of Securing, Deploying and Testing the Application
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
42	Securing, Deploying and Testing the Application (Hands-on Practice)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Practical session, Hands-on coding exercise, Power Point Presentation (PPT)
		CLO	4

		SLO	Understand the concepts of Securing, Deploying and Testing the Application (Hands-on Practice)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
43	API Documentation and Testing Tools (Postman, Swagger)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Demonstration, Hands-on practice, Power Point Presentation (PPT)
		CLO	1
		SLO	Understand the concepts of API Documentation and Testing Tools (Postman, Swagger)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
44	Cloud Deployment of Web Applications (AWS/Azure/Heroku)	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress 2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
		Reading (Reference)	1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
		Pedagogy	Demonstration, Open discussion, Power Point Presentation (PPT)
		CLO	4
		SLO	Understand the concepts of Cloud Deployment of Web Applications (AWS/Azure/Heroku)
		URL	https://lms.nirmauni.ac.in/course/view.php?id=10674
45	Capstone Project: End-to-End Full	Text Book	1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress

	Stack Web Application		2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
	Reading (Reference)		1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly 2. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt. 3. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB 4. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace 5. Hugo Di Francesco, Professional JavaScript, Packt 6. Shama Hoque, Full-Stack React Projects, Packt
	Pedagogy	Project-based learning, Mentoring session	
	CLO	1,2,3,4	
	SLO	Understand the concepts of Capstone Project: End-to-End Full Stack Web Application	
	URL	https://lms.nirmauni.ac.in/course/view.php?id=10674	

f. References

1. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress
2. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress
3. David Flanagan, JavaScript: The Definitive Guide, O'Reilly
4. Juha Hinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt.
5. Shagun Bakliwal, Hands-on Application Development using Spring Boot, BPB
6. Hassan Djirdeh, Nate Murray, and Ari Lerner, Fullstack Vue: The Complete Guide to Vue.js, Createspace
7. Hugo Di Francesco, Professional JavaScript, Packt
8. Shama Hoque, Full-Stack React Projects, Packt

Some More References:

1. <https://www.tutorialspoint.com/>
2. <https://www.javatpoint.com/>
3. <https://www.w3schools.com/>

g. Instructions

Students are expected to carry out assigned work under the Continuous Evaluation (CE) component and LPW component independently. Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of the corresponding component will be affected proportionately in such cases. Turnitin software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.



CERTIFICATE

This is to certify that

Mr. / Ms. _____

Roll No. _____ *of Semester* _____

has satisfactorily completed the course in

Within the four walls of the
Institute.

Date of Submission: _____

Sign In-charge

Head of the Department

Practical List

Sr. No.	Practical Title	Hours	CLO
1	I. Demonstrate JavaScript code embedded within the HTML file II. Install Node.js: Download and install Node.js from nodejs.org. III. create a “script.js” file to print “Hello world” on the server side IV. create an “app.js” file to create and store JSON Object in the File V. Identify the application title that will be used to perform Practical 2 to 10	02	1
2	I. Create a JSON object and display the same using node.js in the terminal. II. Create a Node.js script to read and display the content of the external JSON file. III. Create and display multi-dimensional JSON Arrays, also demonstrate Accessing individual elements of the multi-dimensional JSON Arrays. IV. Create a web-based application example using JavaScript and JSON to manipulate JSON data. In this example, create a simple to-do list application where tasks are stored in a JSON array. Users can add tasks, mark them as completed, and remove them from the list.	02	1
3	I. Create a responsive web page involves using the viewport <u>meta</u> tag in the head of your HTML document. II. Create a responsive web page by using the width and max-width Properties and Changing <u>images</u> according to the browser width. III. Create a responsive web page that demonstrates responsive text size using the "vw" (viewport width) unit in CSS IV. Create a responsive web page that demonstrates <u>media queries</u> for creating breakpoints in the web page.	02	3
4	I. Implement Geolocation API to retrieve the user's current location. II. Implement JavaScript to interact with the Local Storage III. Demonstrating the Drag and Drop API	02	2
5	I. Demonstrate the implementation of Bootstrap online (using the CDN link). II. Demonstrate the implementation of Bootstrap offline. III. Create a simple web page using Bootstrap 5 with HTML for student registration The web page must include the title of the page, roll number, first name, mobile number, email ID, and address.	02	3
6	Demonstrate Tailwind CSS through HTML web pages.	04	3
7	Demonstrate how to create a custom directive and how to use it through HTML web pages (Vue JS). I. Create a custom directive to transform text to uppercase when clicked. II. Creating a Dynamic List with Vue.js Custom Directive III. Create a directive that formats and displays dates in a human-readable format	04	4
8	I. Demonstrate Node.js using MySQL to perform the below operations: Create a database, create a table, insert a record, and update the record. II. Demonstrate Node.js using MySQL to perform the below operations using the user-defined function: delete record, select record, select unique, and drop table.	04	1
9	Demonstrate a Role-Based Authentication and Access Control System using JavaScript.	04	4
10	Demonstrate Node.js using MongoDB to perform Create, Read, Update, and Delete operations.	04	4
11*	Deploy the application and manage version control using Git and GitHub.	-	4
	Total	30	

Nirma University
Institute of Technology
Computer Science and Engineering Department

Practical Policy

B.Tech. Semester - II

Academic Year: 2023-24

Course Code & Name	:	2CS201-Full Stack Web Development
Credit Details	:	Lectures-2, Tutorial-0, Practicals-2, Credits-3
Course Co-ordinator	:	Dr. Devendrasinh Vashi
Contact No. & Email	:	079-71652539, devendra.vashi@nirmauni.ac.in
Office	:	Mon to Fri: 11:00 am to 6:00 pm, Sat: 9:00 am to 3:00 pm
Course Faculty:		
<u>Dr Devendra Vashi</u> Email: devendra.vashi@nirmauni.ac.in , Contact No. 8320092595, Office: E-block, Fifth floor (501).		
<u>Dr. Sunil Gautal</u> , W-Block, 5 th floor		
<u>Dr. Kanu Patel</u> , W-Block, 5th floor		
<u>Dr Somenath Bera</u>		
<u>Prof. Ritiksha Modi</u>		
<u>Prof. Jay Padhiyar</u>		
Common Visiting Hours: Working Days- 1.30 p.m. to 02.00 p.m.		

Rubrics for Regular Evaluation Weightage (0.6)	Rubrics for Final Examination Weightage (0.4)
Active involvement in different tasks related to practical / Eagerness to learn	Demonstration of accurate understanding of the objective of practical
Regularity in attending practical sessions	Fundamental concepts and technical know-how about practical
Preparedness for practical session	Correlation of theoretical concepts with real-life applications
Ability to work in a team	Question-answers, writing and presentation skills
Originality, completeness, presentability and timely submission of assigned work/laboratory manual	Use of appropriate procedures, tools and techniques to conduct experiments and collect data
Discipline during practical session	

Rubric for continuous assessment of practical (10 marks)

Note: Copying in any form is not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Academic integrity is expected from students in all components of course assessment.

Criteria	Satisfactory (0.5 mark)	Adequate (1 mark)	Proficient (1.5 mark)	Outstanding (2 mark)
Lab participation (02 mark)	Student arrive late in the lab and rarely participate in lab proceedings. No attempt/desire to learn.	The student has a tendency to arrive late and unprepared. Such unpunctuality or unpreparedness makes it impossible to fully participate.	The student arrives on time to the lab, but may be unprepared and hence could not participate to the full extent.	The student arrives on time with complete preparation and participates in lab proceedings with full enthusiasm. Even eager to explain concepts to fellow batch mates and ready to assist them.
Definition of objectives and scope (02 mark)	The student is unaware of the practical objectives and concepts	The student has a difficulty in understanding/explaining key concepts of the practical	The student has a basic knowledge of content but may lack understanding of some of the concepts.	The student demonstrates an accurate understanding of the objectives and concepts.
Proper use of procedures (02 mark)	Unable to follow the instructions and performs the experiment.	Follows a limited set of instructions and performs the experiment half-heartedly.	Performs the experiment in a proper manner, however occasionally not following the procedures.	Follows all the instructions given by the instructor and performs the experiment in a perfect manner. Also, influence/emphasise others to follow the procedures.
Result analysis and discussion/ Timely Completion of the work (02 mark)	Calculations/ Graphs/Quizzes are not complete and not submitted within the given deadline.	Calculations/Graphs/Quizzes are partially complete in a very random/ haphazard or disorganised manner. Work is inaccurate and has a number of errors.	Calculations/Graphs/Quizzes are complete. However, student could do the work more neatly by incorporating all the required information.	Calculations/Graphs/Quizzes are complete and neat. They include all the required details like titles, sketches, units etc. Errors, if any are minimal
Question-Answer/Pre presentation (02 mark)	Unable to answer the questions. Poor language and communication with a number of mistakes.	Answers to the questions are basic and superficial suggesting that concepts are not fully grasped. Language and communication is not clear and fluent, suggesting scope for improvement	Questions are answered fairly well barring a few questions. The language is good. Communication is clear.	All the questions are answered completely and correctly. Language is error-free. Communication is clear and fluent. No grammatical mistakes.

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Sr. No.	Title of experiment	Date	Marks (as per rubrics)					Total Marks (10)	Sign
			2	2	2	2	2		
1.	Demonstration of JavaScript, Node.js and JSON Object through HTML webpages.								
2.	Demonstration of Node.js, JSON Arrays and JSON Object through HTML webpages.								
3.	Create responsive web pages using the width and max-width Properties, use of media queries and the “vw” unit.								
4.	Demonstrate Geolocation API, Local Storage and Drag and Drop API through HTML webpages.								
5.	Demonstrate the implementation of Bootstrap online and offline mode through HTML webpages.								
6.	Demonstrate Tailwind CSS through HTML web pages.								
7.	Demonstrate how to create a custom directive and how to use it through HTML web pages.								
8.	I. Demonstrate Node.js using MySQL to perform the below operations: Create a database, create a table, insert a record, and update the record. II. Demonstrate Node.js using MySQL to perform the below operations: delete record, select record, select unique, drop table.								
9.	Demonstrate a Role-Based Authentication and Access Control System using JavaScript.								
10.	Performs Create, Read, Update, and Delete operations using MongoDB.								

Lab Faculty Signature and Name